BTVNANOCOMMISSIONING · ZB_ZC_RATIO WORKFLOW

# Z(→ℓℓ) + b/c jets in Run 3: first look at data

## Summer23

2023 C data vs MC, a small sample (0.6 $fb^{-1}$ in Z→μμ, 0.6 $fb^{-1}$ in Z→ee), AK4 jets

26 October 2026

# What is new since last time

**DATA**

## 2023 C

First look at data. Muon and EGamma, 22Sep2023 reprocessing.

**CAMPAIGN**

## Summer23

One of four campaigns run (Summer22, Summer22EE, Summer23, Summer23BPix). Each has its own corrections and its own deck; they are not combined.

**CORRECTIONS**

## Full set

Jet energy corrections, jet veto maps, lepton energy scale, lepton and trigger SFs, all from the official POG files. Hsin-Wei's latest code merged in.

This is a small, quick run: 20 files per MC sample and 20 data files per channel and campaign, run interactively. The aim is to see whether data and MC agree before we scale up. The selection is the same as in the MC-only study: two leptons on the Z peak and at least one AK4 jet with pT > 30 GeV. No b- or c-tag requirement yet.

# Data used

| Dataset (DAS) | Files | Lumisections | brilcalc [pb$^{-1}$] | Factor | Used [pb$^{-1}$] |
|---|---|---|---|---|---|
| /EGamma0/Run2023C-22Sep2023_v1-v1/NANOAOD | 19 | 3,588 | 1133 | 0.5 | 566 |
| /Muon0/Run2023C-22Sep2023_v1-v1/NANOAOD | 20 | 3,526 | 1210 | 0.5 | 605 |

## Summer23: Z→μμ 605 pb$^{-1}$, Z→ee 566 pb$^{-1}$

The 22Sep2023 re-reconstruction in NanoAODv12, 20 files per channel spread over the eras. Z→μμ comes from the Muon PD (IsoMu24), Z→ee from the EGamma PD (Ele30_WPTight). Muon and EGamma files cover different lumisections, so each channel has its own luminosity and the MC is scaled to it.

In 2023 each PD is split in two halves (Muon0/Muon1, EGamma0/EGamma1) that share every lumisection, each half holding about 50% of the events. We use only the first half and count half of its luminosity (factor 0.5).

EGamma0 Run2023C: one file timed out when opened and was skipped. Its lumisections are not counted either, so the luminosity stays consistent.

# Golden JSON and luminosity

**GOLDEN JSON**

### Cert_Collisions2023_366442_370790_Golden.json

The official 2023 golden JSON, read from the CMS central area on cvmfs (cms-griddata …/DC/Collisions23/latest). This is the newest version, after the 2026 tracker re-review.

**LUMINOSITY**

### brilcalc, normtag_PHYSICS

Run in the BRIL container on LPC, on exactly the lumisections we processed. Muon and EGamma get their own numbers.

**Checked against the newest golden JSON**

All 6,868 processed lumisections are golden in the newest version.

# MC samples

| Sample | Generator | Run3Summer23NanoAODv12-… | σ [pb] | Files |
|---|---|---|---|---|
| DYto2L-2Jets_MLL-50 | amcatnloFXFX-pythia8 | 130X_mcRun3_2023_realistic_v14-v2 | 6346 | 20 / 506 |
| DYto2L-2Jets_MLL-50_PTLL-100 | amcatnloFXFX-pythia8 | 130X_mcRun3_2023_realistic_v15-v2 | 105 | 19 / 815 |
| DYto2L-2Jets_MLL-50_PTLL-200 | amcatnloFXFX-pythia8 | 130X_mcRun3_2023_realistic_v15-v2 | 11.24 | 20 / 398 |
| DYto2L-2Jets_MLL-50_PTLL-400 | amcatnloFXFX-pythia8 | 130X_mcRun3_2023_realistic_v15-v3 | 0.6433 | 20 / 257 |
| DYto2L-2Jets_MLL-50_PTLL-600 | amcatnloFXFX-pythia8 | 130X_mcRun3_2023_realistic_v15-v2 | 0.09941 | 19 / 158 |
| TTto2L2Nu | powheg-pythia8 | 130X_mcRun3_2023_realistic_v14-v2 | 98.88 | 20 / 198 |
| WW | pythia8 | 130X_mcRun3_2023_realistic_v14-v2 | 75.9 | 20 / 125 |
| WZ | pythia8 | 130X_mcRun3_2023_realistic_v14-v2 | 27.6 | 20 / 60 |
| ZZ | pythia8 | 130X_mcRun3_2023_realistic_v14-v2 | 12.17 | 20 / 11 |

Full DAS name: /<sample>_TuneCP5_13p6TeV_<generator>/Run3Summer23NanoAODv12-<processing>/NANOAODSIM. Files: used / available. The DY pT($\ell\ell$) samples are stitched: each binned sample keeps only its own LHE pT($\ell\ell$) slice and the inclusive one covers pT($\ell\ell$) < 100 GeV. DY events are split by the hadron flavour of the leading jet (b, c or light). $t\bar{t}$ and dibosons are not split. Cross sections as in the MC-only study. DYto2L-2Jets_MLL-50_PTLL-100, DYto2L-2Jets_MLL-50_PTLL-600: one file timed out when opened and was skipped; the normalisation uses the files that were read.

# Corrections and scale factors

| Correction | Applied to | POG file | Key / tag |
|---|---|---|---|
| Jet energy corrections | data + MC | JME jet_jerc | Summer23Prompt23_V4 (data: with residuals) |
| Jet energy resolution | MC | JME jet_jerc, jer_smear | Summer23Prompt23_RunCv1234_JRV3 |
| MET | data + MC | from the corrected jets | type-1 PuppiMET |
| Jet veto map | data + MC | JME jetvetomaps | Summer23Prompt23_RunC_V1 |
| Muon momentum scale and resolution | data + MC | MUO muon_scalesmearing | Run 3 recipe |
| Electron energy scale / smearing | data / MC | EGM electronSS_EtDependent | Scale / SmearAndSyst |
| Pileup weight | MC | LUM puWeights | Collisions2023_366403_369802_eraBC_GoldenJson |
| Muon ID and isolation SF | MC | MUO muon_Z | NUM_TightID_DEN_TrackerMuons, NUM_TightPFIso_DEN_TightID |
| Muon trigger SF (leading muon) | MC | MUO muon_Z | NUM_IsoMu24_DEN_CutBasedIdTight_and_PFIsoTight |
| Electron ID and reco SF | MC | EGM electron | Electron-ID-SF: Tight and Reco (2023PromptC) |
| Electron trigger SF (leading electron) | MC | EGM electronHlt | HLT_SF_Ele30_TightID |
| b- and c-tag SF | MC | BTV btagging, ctagging | not applied yet (no tag cut) |

Every file is the official one on cvmfs: /cvmfs/cms-griddata.cern.ch/cat/metadata/<POG>/Run3-23CSep23-Summer23-NanoAODv12/latest/. The framework uses them for all BTV workflows.

# POG recommendations vs what we do (1/2)

## Data and jets

| Topic | POG recommendation | What we do | Status |
|---|---|---|---|
| Golden JSON (DC) | Use the latest golden JSON | Latest version from cvmfs | **Same** |
| Luminosity (LUM) | brilcalc with normtag_PHYSICS | Same, on the lumisections we processed | **Same** |
| Pileup (LUM) | Official pileup weights | Same file | **Same** |
| Jet energy corrections (JME) | Summer23Prompt23_V4, resolution Summer23Prompt23_RunCv1234_JRV3 | Same tags | **Same** |
| Jet ID (JME) | Tight lepton veto ID; the NanoAODv12 jetId branch needs the JME fix | Fix applied | **Same** |
| Jet veto map (JME) | Veto the event if a jet with pT > 15 GeV, tight ID, EM fraction < 0.9 and away from muons falls in the map | Framework vetoes on any jet in the map. Drops about 6% of Z→μμ events instead of 1–2%; data/MC moves by 0.3% or less | **Differs** |
| Rare high-pT jets (JME) | Reject jets with rawFactor > 0.9 (known JEC issue) | Not applied. About 1 event in a million, mostly forward, outside $\|\eta\| < 2.5$ | **Not yet** |

Read on 26 October 2026: BTV scale factor page, MUO usage recommendations, JME jet energy scale page, EGamma Run 3 TWiki (last edited 2023, older than the cvmfs files we use). Jet veto recipe from the JME jet veto map recommendation. The 6% and 1–2% are quick estimates on one chunk.

# POG recommendations vs what we do (2/2)

## Leptons and tagging

| Topic | POG recommendation | What we do | Status |
|---|---|---|---|
| Muon ID and isolation (MUO) | Tight ID, PF isolation < 0.15 (Tight), SFs from muon_Z | As recommended | **Same** |
| Muon trigger SF (MUO) | IsoMu24 SF per muon (Tight ID + Tight iso) | Leading muon only (pT > 35 GeV, on the plateau). Effect of using both muons not checked yet | **Simplified** |
| Electron ID and reco (EGM) | Cut-based Tight ID, SFs for the 22Sep2023 re-reco | As recommended | **Same** |
| Electron trigger SF (EGM) | HLT_SF_Ele30_TightID per electron | Leading electron only. Using both electrons raises the Z→ee MC by 1.1–1.9% (quick estimate) | **Simplified** |
| Lepton energy scale (EGM, MUO) | Scale in data, smearing in MC | Same files and recipe. The Z→ee peak is 1.5–4% wider in MC than in data (all four campaigns), to be understood | **Check** |
| b/c tagging (BTV) | AK4 PUPPI, pT > 20, \|η\| < 2.5, tight ID, hadronFlavour. PNet b: fixed-WP and shape SFs; PNet c: fixed-WP SFs only | pT > 30, \|η\| < 2.5, tight ID, hadronFlavour. No tag SFs yet (no tag cut) | **Not yet** |
| L1 prefiring | No prefiring weight in Run 3 | None | **Same** |

c-tagging has no shape SFs for 2022/2023, only fixed working points. A template fit in the CvL–CvB plane needs a plan for that. UParT SFs exist only for NanoAODv15.

# What Hsin-Wei's update brought

We merged her latest coffea_machine branch (5 commits, up to 1 October) into ours on 26 October, before this run.

FIXED FOR EVERYONE

## Electron SFs with two electrons

The shared electron SF code used to keep only the last electron's SF. Her fix multiplies the SFs of all electrons. We had worked around it by calling it once per electron, so our numbers do not change, but the common code is now right.

SAFER

## Empty lepton slots

The SF code now fills empty lepton slots before calling the POG files. The 2023 reco SFs also need the electron φ, and that path is now protected too.

TO REUSE

## Matched electron trigger SF

Her Run 2 electron trigger SF takes the electron that fired the trigger, matched to the HLT object. That is the right way to do it, and we plan to bring it to Run 3 instead of using the leading electron.

### Bottom line

Her Run 2 electron SF files and settings do not touch the Run 3 inputs, and both branches now share one correction code. Where files clashed we kept our validated Run 2 selection. One thing to tell her: in the muon SF code the trigger-SF pT clip gets overwritten (we do not use that path).

# Cutflow: data vs MC

| | trigger | leptons | OS | Zmass | MET | jet |
|---|---|---|---|---|---|---|
| Z→ee data | 6,462,593 | 169,655 | 168,160 | 160,396 | 152,132 | 27,942 |
| MC | 587,415 | 169,260 | 168,186 | 161,040 | 150,025 | 28,476 |
| data / MC | 11.00 | 1.00 | 1.00 | 1.00 | 1.01 | 0.98 |
| Z→μμ data | 7,329,698 | 335,564 | 335,406 | 317,896 | 302,493 | 54,543 |
| MC | 802,086 | 338,231 | 338,187 | 324,832 | 304,818 | 56,670 |
| data / MC | 9.14 | 0.99 | 0.99 | 0.98 | 0.99 | 0.96 |

Data/MC only means something from the two-lepton step on. At the trigger step the data are mostly single-lepton events (W→ℓν, multijet) that fire the same trigger but are not simulated: we only have DY, $t\bar{t}$ and VV. The two-lepton cut keeps 2–5% of the triggered data but 30–40% of the MC, and data/MC drops from about 10 to about 1. MC here is before SFs (generator weight × σ × L / Σw only), so data/MC is expected a few % off 1 at the lepton steps.

# Yields after the full selection

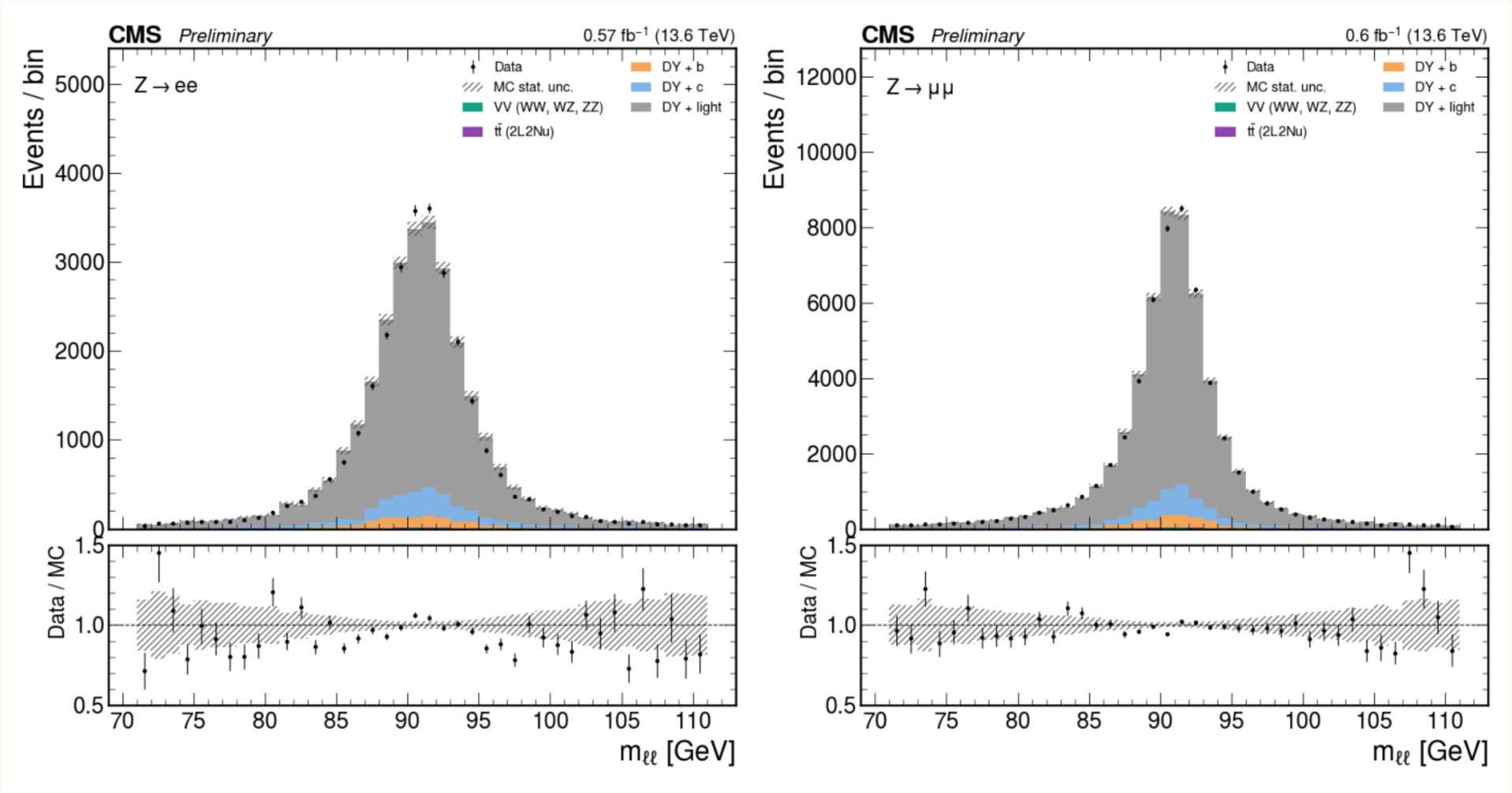
| | DY + light | DY + c | DY + b | tt̄ | VV | MC | Data | Data/MC |
|---|---|---|---|---|---|---|---|---|
| Z→ee | 24,955 | 2,456 | 1,102 | 103 | 92 | 28,709 | 27,942 | 0.973 |
| Z→μμ | 48,009 | 4,663 | 2,228 | 207 | 174 | 55,282 | 54,543 | 0.987 |

Z→ee: DY + b is 3.9% and DY + c is 8.6% of DY (c/b = 2.23); tt̄ + VV is 0.7% of the MC.

Z→μμ: DY + b is 4.1% and DY + c is 8.5% of DY (c/b = 2.09); tt̄ + VV is 0.7% of the MC.

# Dilepton mass

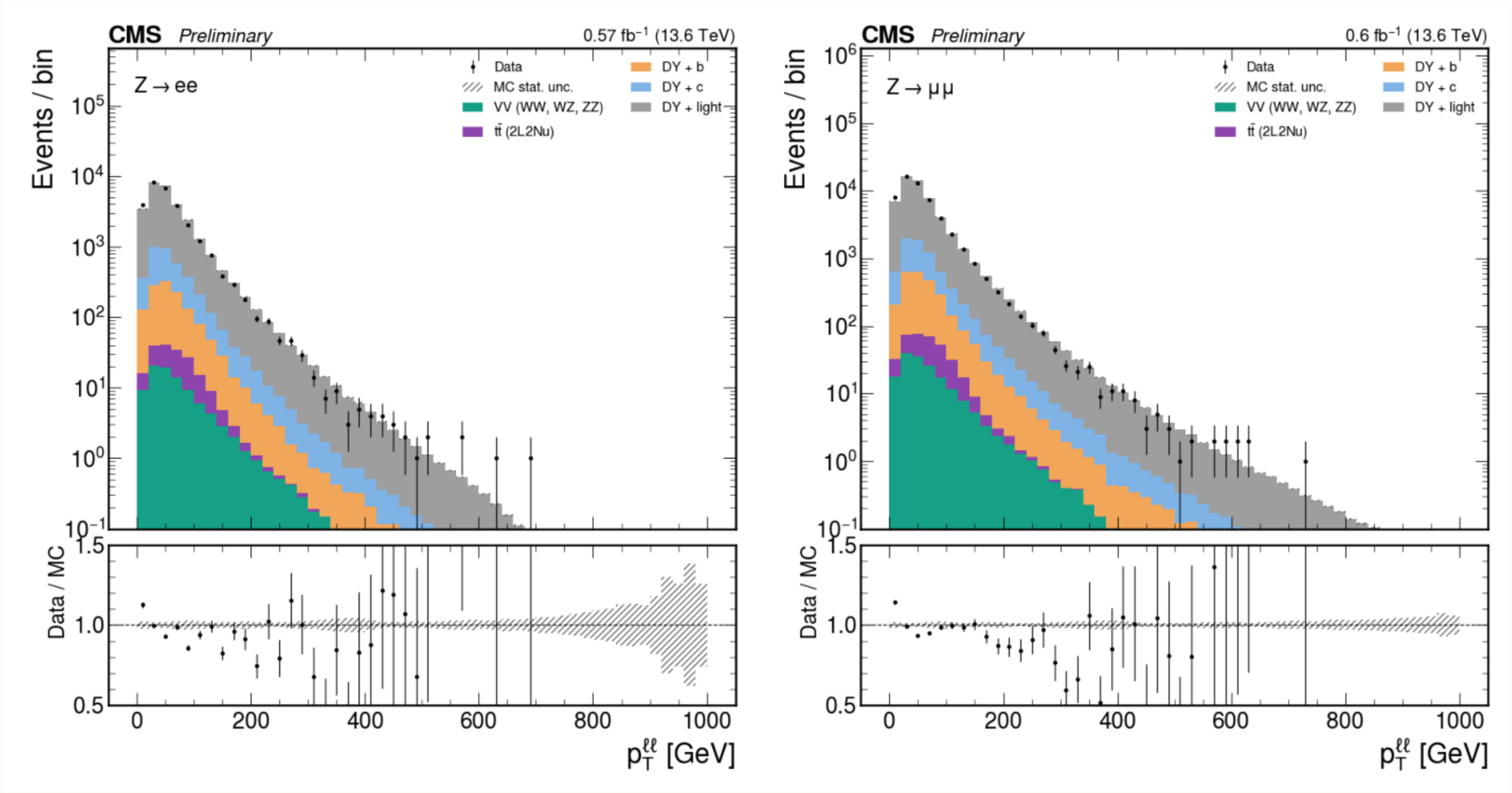
The Z peak in data and MC. Linear y-axis.




Run 3, 13.6 TeV · 2023 C · 20 files per MC sample and per data channel · AK4 jets · Z→ee left, Z→μμ right · band: MC stat. uncertainty · 12

# Dilepton pT

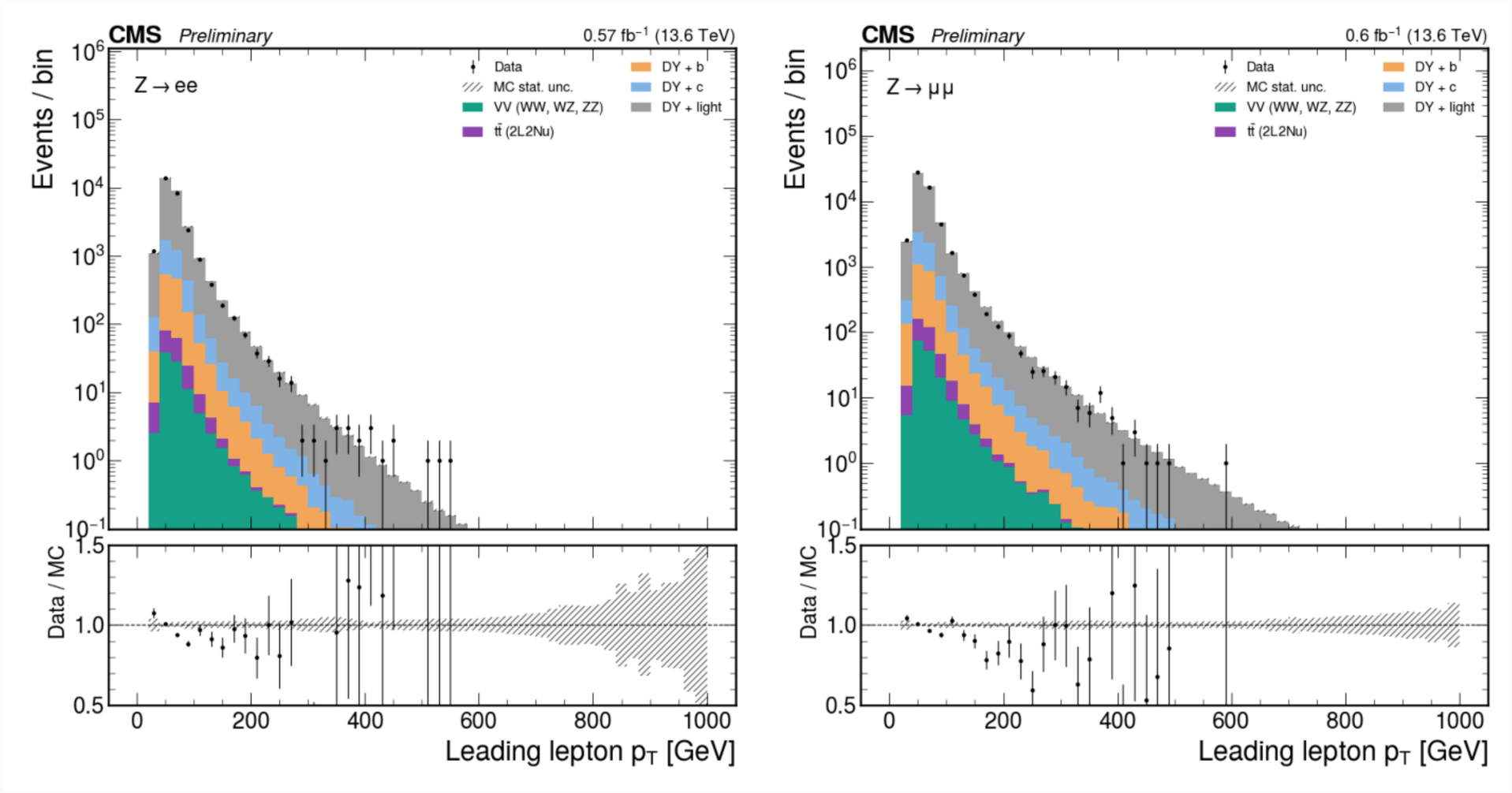
pT of the Z candidate. No cut on it.




Run 3, 13.6 TeV · 2023 C · 20 files per MC sample and per data channel · AK4 jets · Z→ee left, Z→μμ right · band: MC stat. uncertainty · 13

# Leading-lepton pT

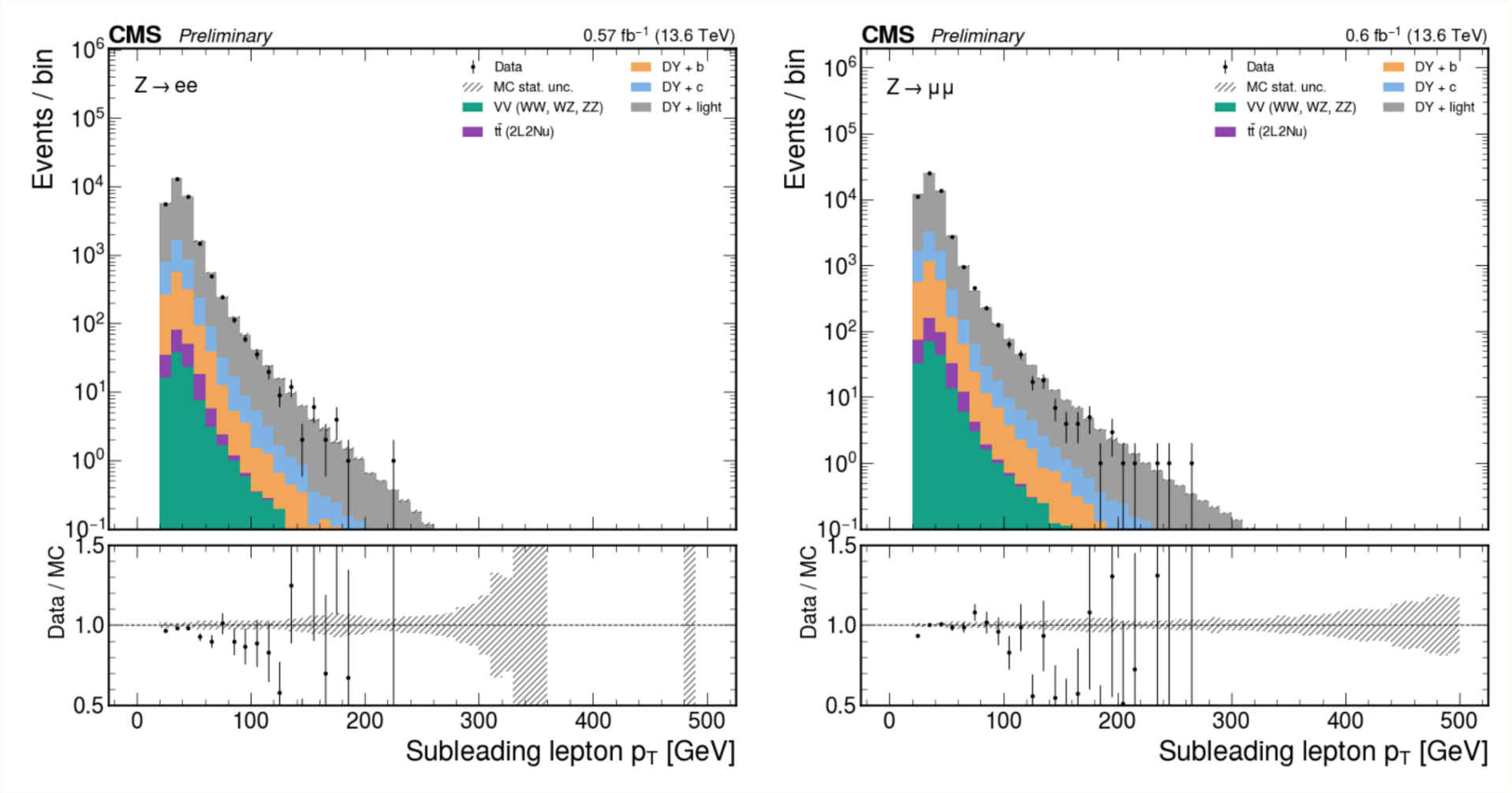
Leading lepton, pT > 35 GeV required.




Run 3, 13.6 TeV · 2023 C · 20 files per MC sample and per data channel · AK4 jets · Z→ee left, Z→μμ right · band: MC stat. uncertainty · 14

# Subleading-lepton pT

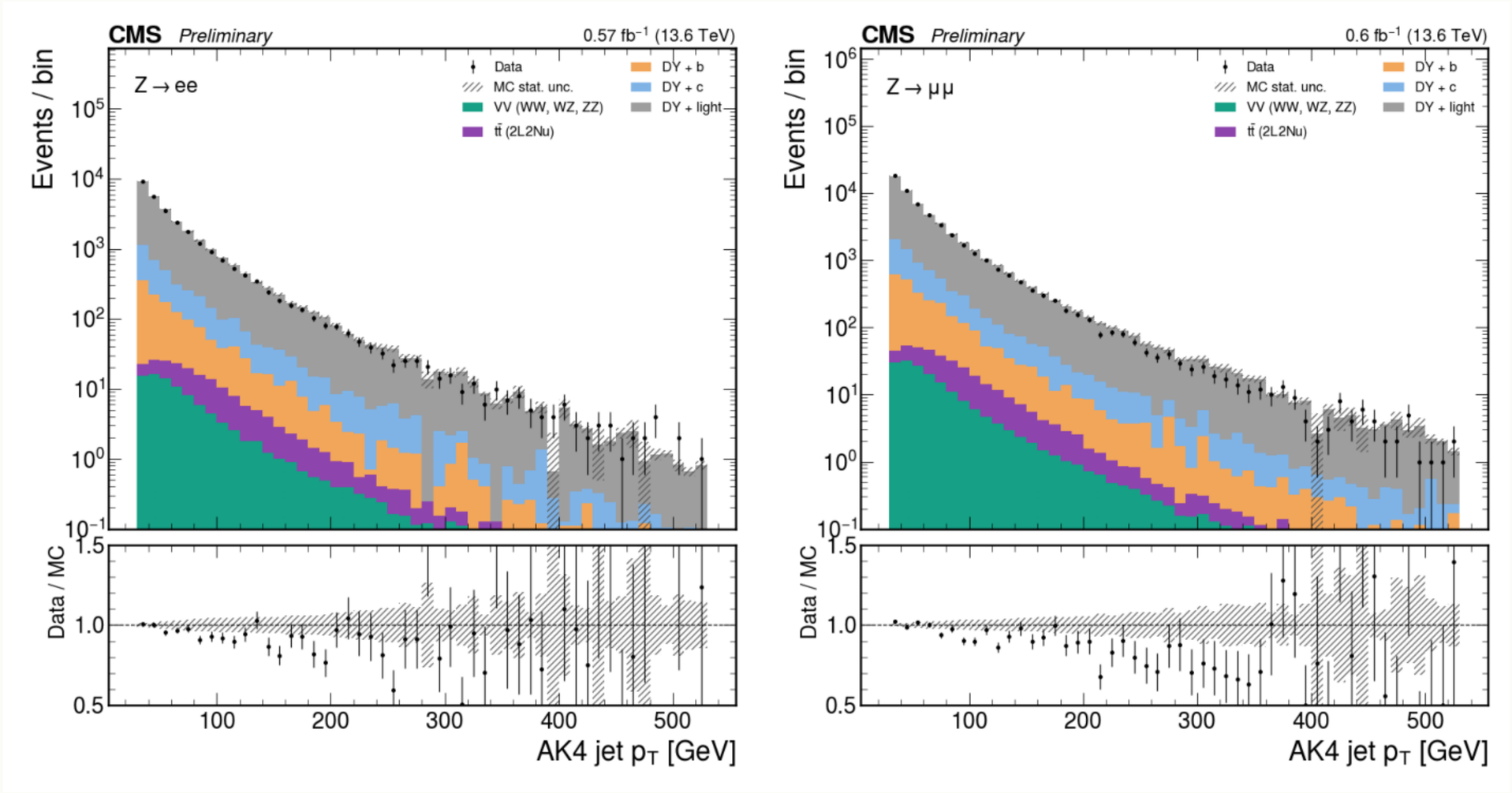
Subleading lepton, pT > 25 GeV required.




Run 3, 13.6 TeV · 2023 C · 20 files per MC sample and per data channel · AK4 jets · Z→ee left, Z→μμ right · band: MC stat. uncertainty · 15

# Leading AK4 jet pT

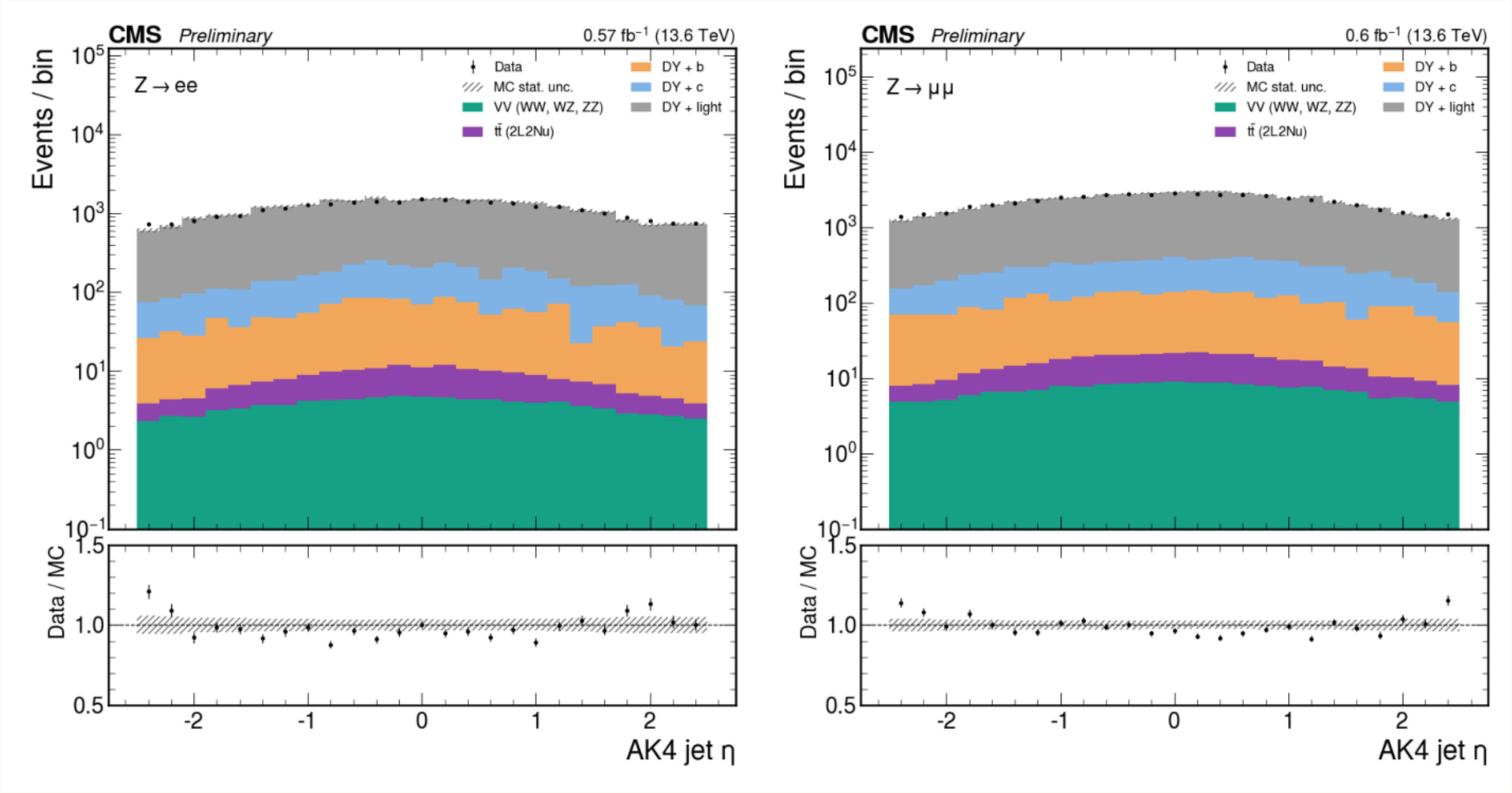
Leading lepton-cleaned AK4 jet, pT > 30 GeV required.




Run 3, 13.6 TeV · 2023 C · 20 files per MC sample and per data channel · AK4 jets · Z→ee left, Z→μμ right · band: MC stat. uncertainty · 16

# Leading AK4 jet η

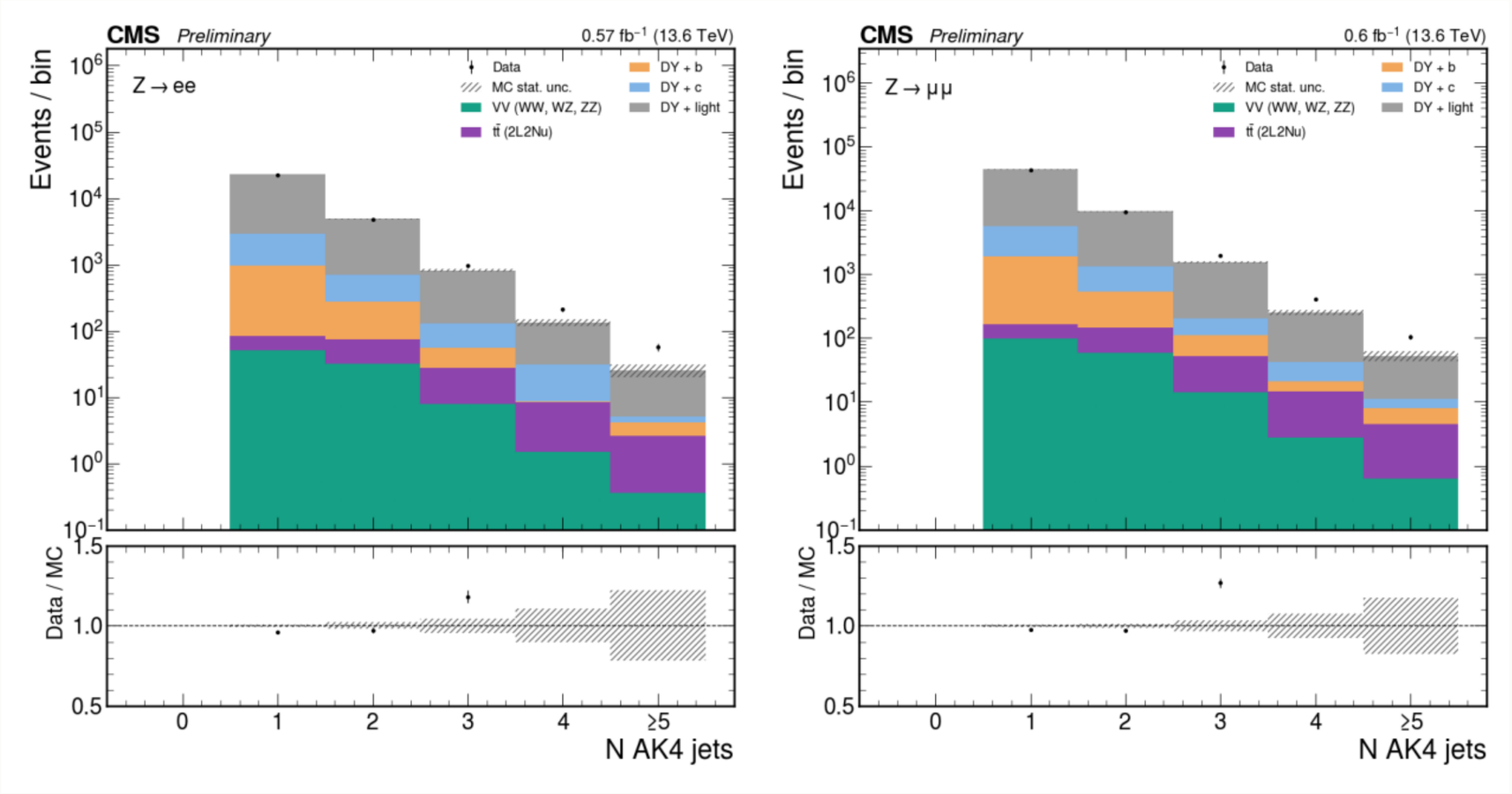
Pseudorapidity of the leading jet, |η| < 2.5.




Run 3, 13.6 TeV · 2023 C · 20 files per MC sample and per data channel · AK4 jets · Z→ee left, Z→μμ right · band: MC stat. uncertainty · 17

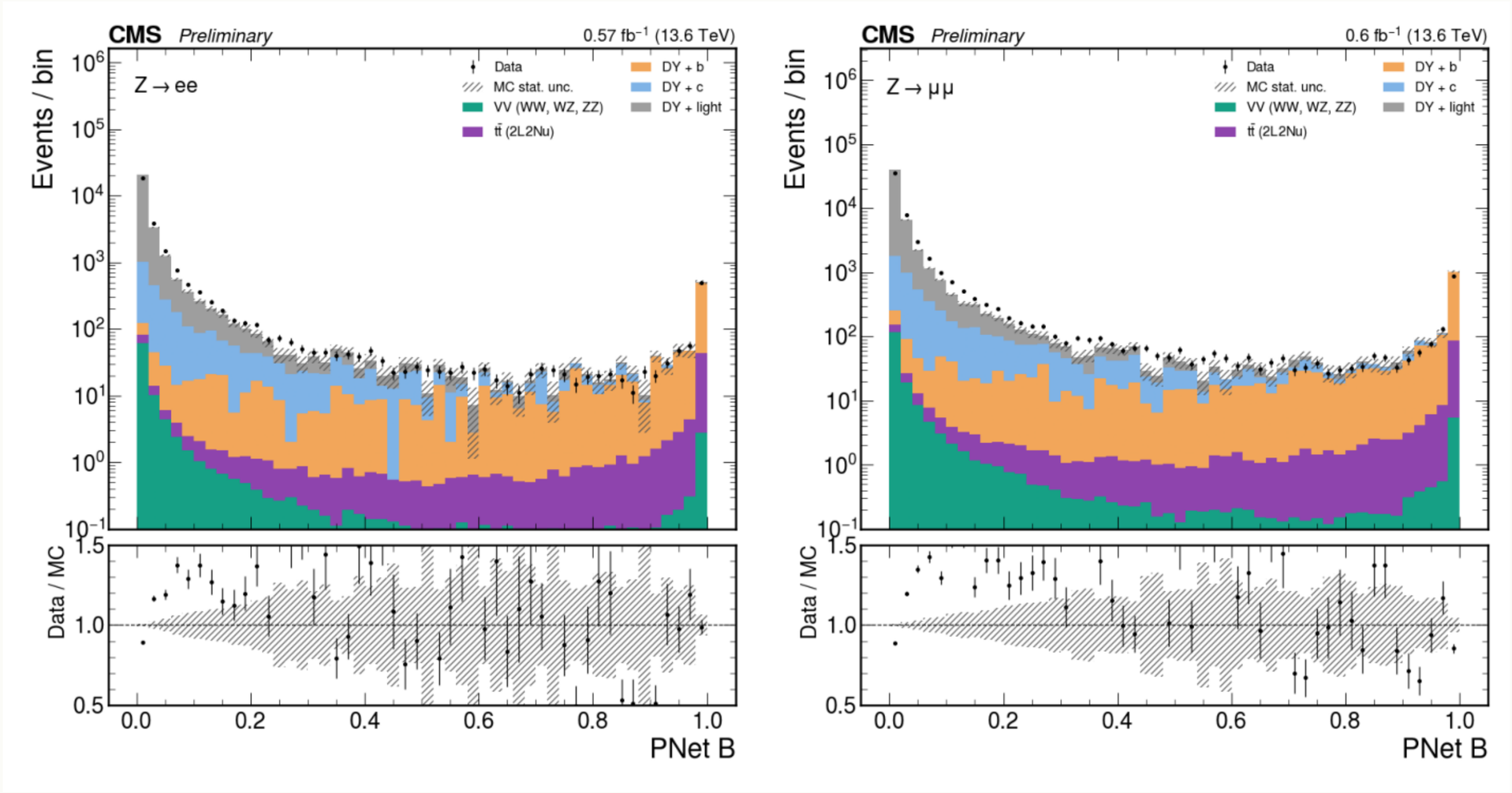
# Number of AK4 jets

Jets with pT > 30 GeV and |η| < 2.5. The last bin holds 5 or more jets. These outputs were filled after the at-least-one-jet cut, so the 0 bin is empty. The workflow now fills it before that cut; the next run will have the 0-jet bin.



# PNet b-tag score

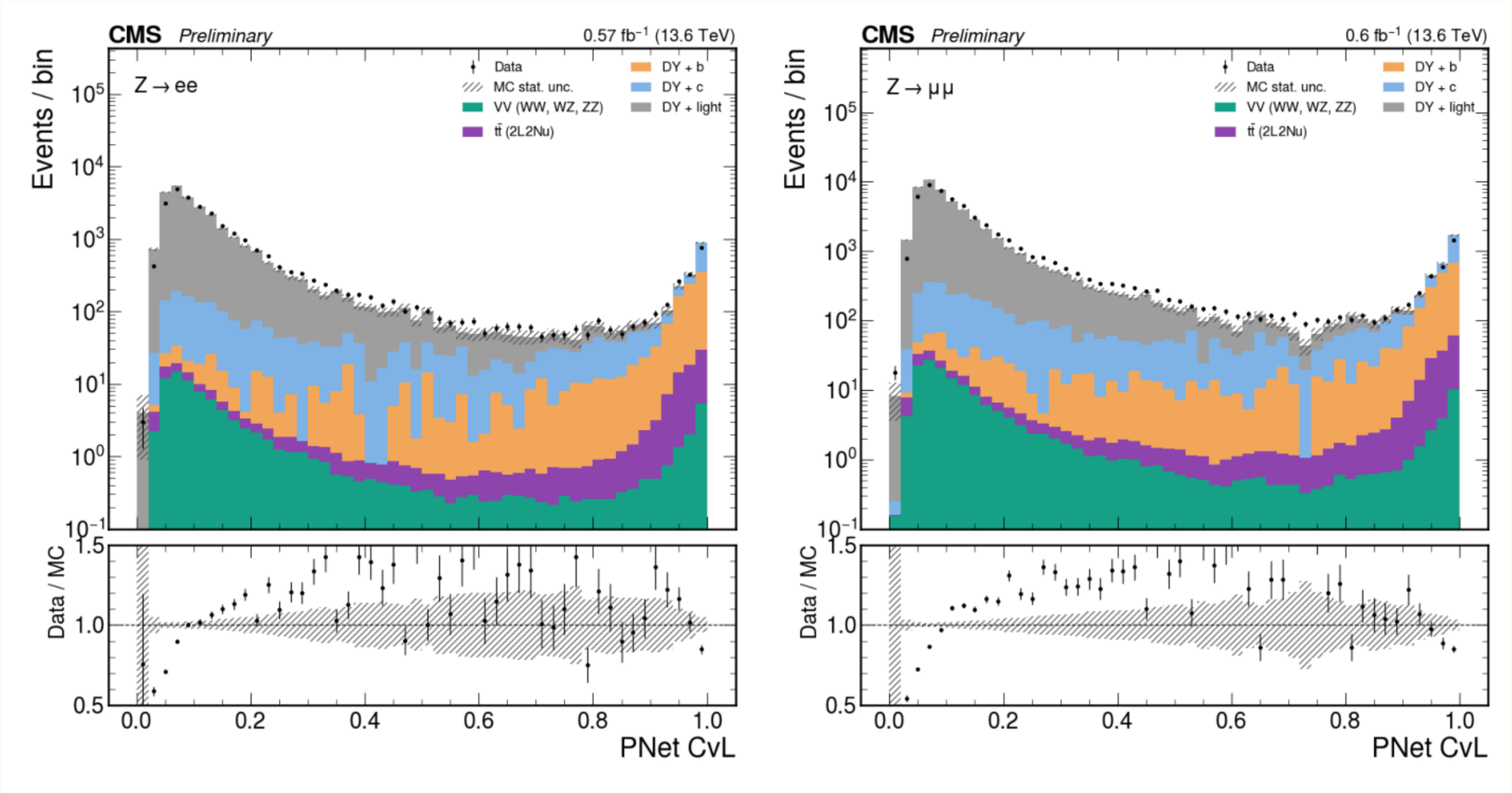
Leading jet. b jets sit near 1, light jets near 0.




Run 3, 13.6 TeV · 2023 C · 20 files per MC sample and per data channel · AK4 jets · Z→ee left, Z→μμ right · band: MC stat. uncertainty · 19

# PNet charm vs light

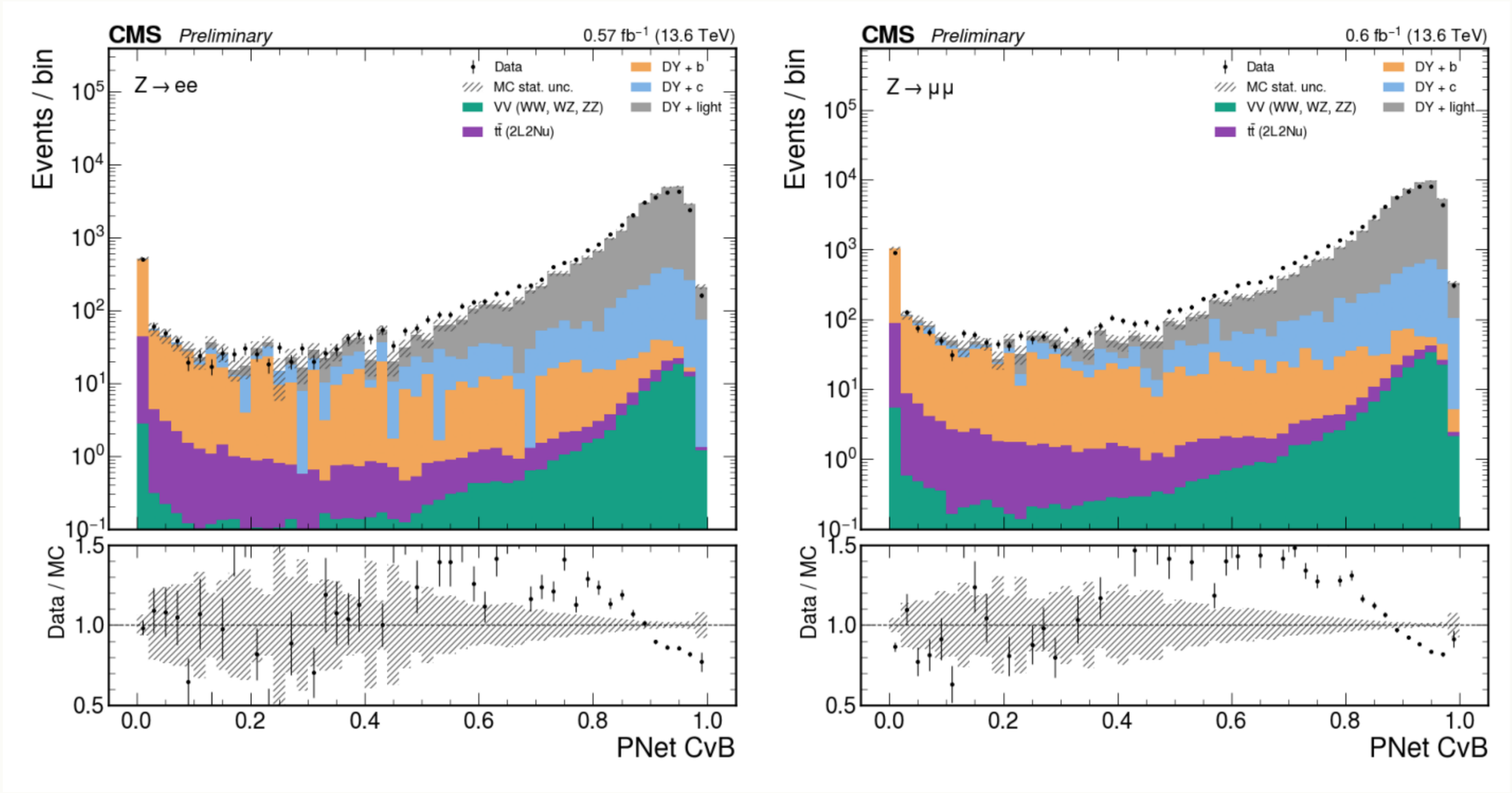
Leading jet.




Run 3, 13.6 TeV · 2023 C · 20 files per MC sample and per data channel · AK4 jets · Z→ee left, Z→μμ right · band: MC stat. uncertainty · 20

# PNet charm vs b

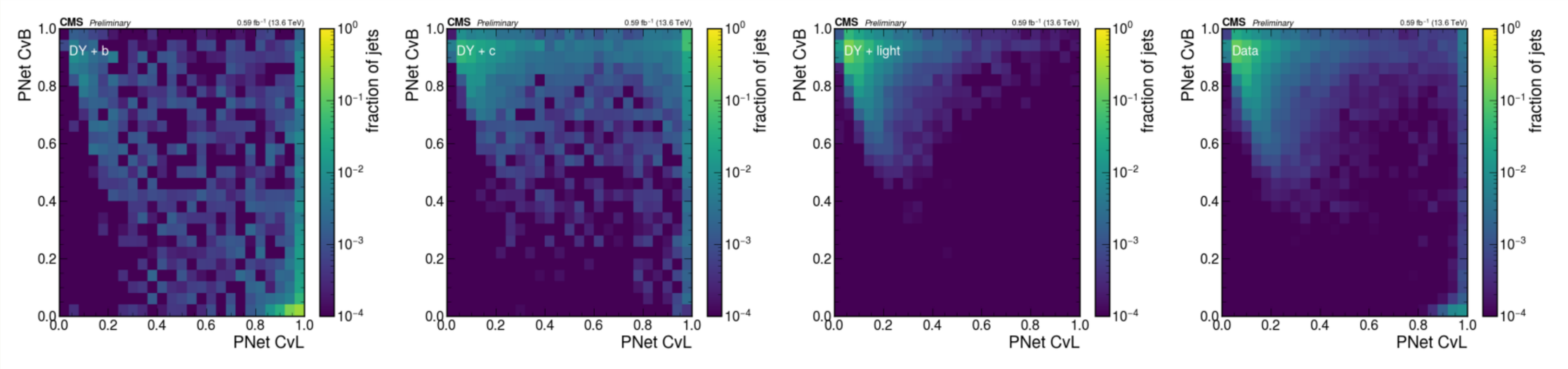
Leading jet.




Run 3, 13.6 TeV · 2023 C · 20 files per MC sample and per data channel · AK4 jets · Z→ee left, Z→μμ right · band: MC stat. uncertainty · 21

# PNet charm-tagging plane: CvL vs CvB

Leading jets, both channels summed, each panel normalized to unit area. Data is a mix of all flavours, mostly light.



# Summary and next steps

**Data and MC agree within 3% in both channels**

With 0.57 $fb^{-1}$ in Z→ee and 0.60 $fb^{-1}$ in Z→μμ (Muon0 and EGamma0, half of their luminosity), data/MC after the full selection is 0.97 ± 0.01 (Z→ee) and 0.99 ± 0.01 (Z→μμ). The Z peak, the jet pT and η and the PNet scores have the same shape in data and MC. DY + b is about 4% and DY + c about 8.5% of the DY events. The half-PD treatment of 2023 works. This campaign is in good shape.

- Full run on condor, after moving to the JME-style jet veto.
- Electron trigger SF from both electrons, matched to the trigger object as in Hsin-Wei's Run 2 code.
- Find out why the Z→ee peak is wider in MC than in data (electron smearing).
- b/c-tag SFs once we cut on the tagger; decide what to do without c-tag shape SFs.